

Wasmir Chowdhury

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Integrated BS/MS in Electrical and Computer Engineering, GPA: 3.21

May 2028

Relevant Coursework

Embedded Systems, Real-Time Operating Systems, Structure and Design of Digital Systems, Computer Systems

WORK EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

WiseLAB (Carnegie Mellon University)

Pittsburgh, PA

- Bringing up two ZED X One GS cameras as a hardware-synced (15 μ s) stereo pair on a Jetson AGX Orin, debugging Linux kernel sensor enumeration issues along the way
- Building a real-time stereo H.265 capture pipeline in GStreamer using Jetson hardware encoder nodes, encoding both streams on-device with no intermediate transcode

Course Teaching Assistant

August 2026 – Present

18-349: Introduction to Embedded Systems (Carnegie Mellon University)

Pittsburgh, PA

- Lead two weekly recitations (5 students each) on embedded architectures and real-time scheduling
- Hold 5 hours of office hours weekly, guiding students through ARM Cortex-M labs on interrupts, memory management, and RTOS development

PROJECT EXPERIENCE

Real-Time Operating System (RTOS) Kernel | C, ARM Assembly

- Built a preemptive RMS scheduler on ARM Cortex-M4 with UB-test-based admission control
- Implemented HLP/IPCP mutex locking in C to prevent priority inversion and deadlocks across concurrent tasks
- Wrote a PendSV context switcher in ARM Assembly to manage thread state and CPU register save and restore
- Validated the scheduler with three concurrent threads (50ms PID control, 500ms LCD, 2s UART), holding speed within 10% of target while meeting every deadline

STM32 Bare-Metal Peripheral Drivers | C, STM32, ARM Cortex-M

- Implemented interrupt-driven I2C and UART drivers from scratch via direct peripheral register configuration
- Debugged hardware interface timing using GDB over JTAG and an oscilloscope to validate driver correctness

STM32 Embedded Car PCB | Fusion360, PCB Design, STM32

- Captured a schematic and routed a two-layer PCB in Fusion360 for an STM32 motor-control board
- Calculated regulator and LED resistor values from datasheets, passed DRC/ERC, and generated Gerbers

QNX Real-Time Vibration Monitor (Hackathon) | Python, QNX RTOS

3rd Place, QNX Track

- Built a real-time vibration monitor on Raspberry Pi 5 with QNX RTOS, sampling an MPU6050 via I2C at 100Hz
- Navigated QNX hardware bring-up: I2C resource manager limits, wake sequencing, GPIO drivers, slog2 logging, POSIX signals

Network-on-Chip Router/Node Design (18-341) | SystemVerilog, VCS

- Designed synthesizable Node and Router FSMs for a packet-switched NoC with an 8-bit-to-32-bit reassembly FIFO
- Implemented round-robin arbitration for fair multi-port routing, verified against VCS testbenches

LEADERSHIP EXPERIENCE

EMERGE

July 2024 – May 2026

Mentor, Carnegie Mellon University

Pittsburgh, PA

- Facilitate monthly group meetings with mentees to offer academic, personal, and professional guidance

TECHNICAL SKILLS

Embedded Systems: STM32, ARM Cortex-M, Computer Architecture, RTOS (RMS, HLP/IPCP), Bare-Metal Programming, Memory Management, I2C, UART, Interrupts, GDB, JTAG, Oscilloscope, Logic Analyzer

Software: C, C++, ARM Assembly, Python, Shell/Bash, SystemVerilog, FPGA, Linux, Data Structures, Git/GitHub, Vivado, Fusion360 PCB